# USER INTERFACE DESIGN

# EXPERIMENT – 2

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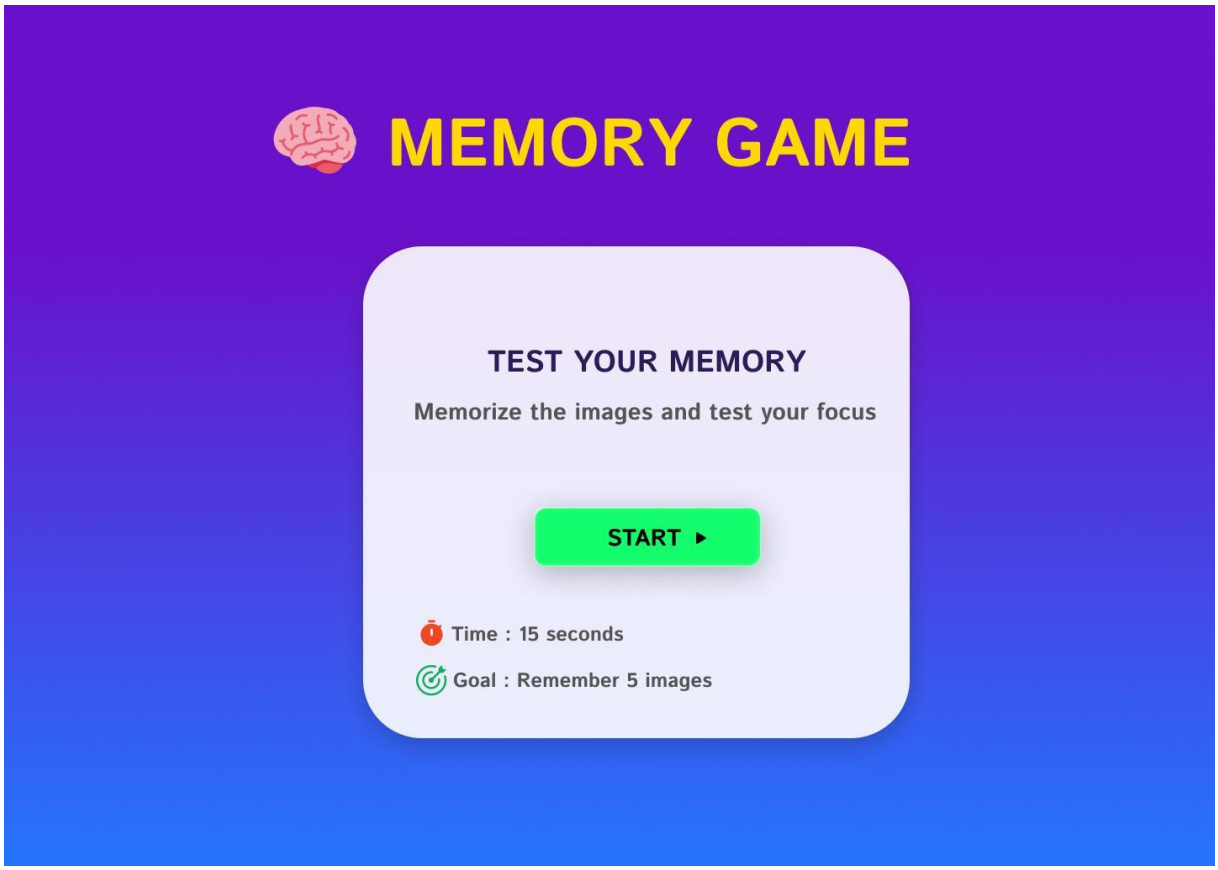
Design a UI where users recall visual elements (e.g., icons or text chunks). Evaluate the effect of chunking on user memory.

## FRAME 1: INSTRUCTION SCREEN




### Purpose of the Screen

This screen introduces the user to the Memory Game and explains the task clearly before starting. It prepares the user mentally for the upcoming memory challenge.

### Key UI Elements

- **Title: “MEMORY GAME”**
  Clearly indicates the purpose of the application.
- **Instruction Card**
  - “Test Your Memory”
  - “Memorize the images and test your focus”
  - Simple, readable content reduces cognitive load.
- **Start Button**
  - Highlighted in green to attract attention.
  - Indicates the beginning of the memory task.
  - Acts as the main call-to-action.
- **Time & Goal Info**
  - Time: 15 seconds
  - Goal: Remember 5 images
    This sets expectations before the game starts.

### UX & Cognitive Benefit

- Instructions are short and direct.
- Visual hierarchy helps users understand what to do quickly.
- The clean layout avoids distraction and improves focus.

## FRAME 2: CHUNKING PHASE (MEMORIZATION SCREEN)

## Purpose of the Screen

This is the **visual encoding phase**, where users observe and memorize a set of images within a limited time.

## Key UI Elements

- **Heading:** “SELECT THE IMAGES YOU REMEMBER”
- **Grid of Fruit Images**
  - Fruits are arranged neatly in rows.
  - Images are visually distinct (apple, banana, kiwi, pineapple, etc.)
  - Helps users form mental groups (chunking).
- **Timer / Progress Indicator**
  - Shows limited viewing time.
  - Creates urgency and forces quick memory processing.

## Chunking Concept Used

Users naturally group images by:

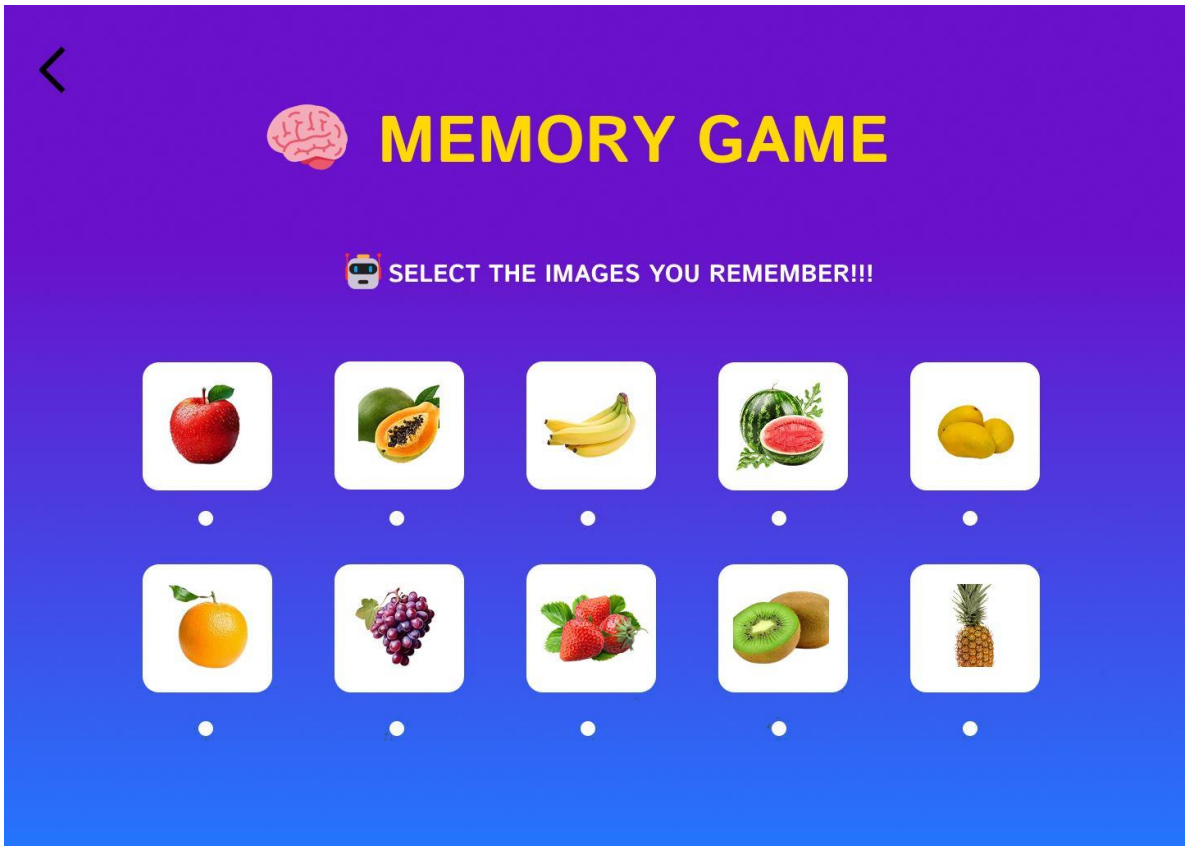
- Fruit type
- Color
- Shape
- Familiarity

This improves short-term memory retention and makes recall easier.

### UX Benefit

- Prevents overload by grouping items visually.
- Improves recall accuracy.
- Engages the user through visual learning.

## FRAME 3: IMAGE SELECTION & VALIDATION SCREEN



## Purpose of This Screen

This screen tests the user's memory by allowing them to select the images they remember from the previous screen.

Unlike traditional quiz systems, **there is no submit button**.
The system evaluates the user's selection **instantly**.

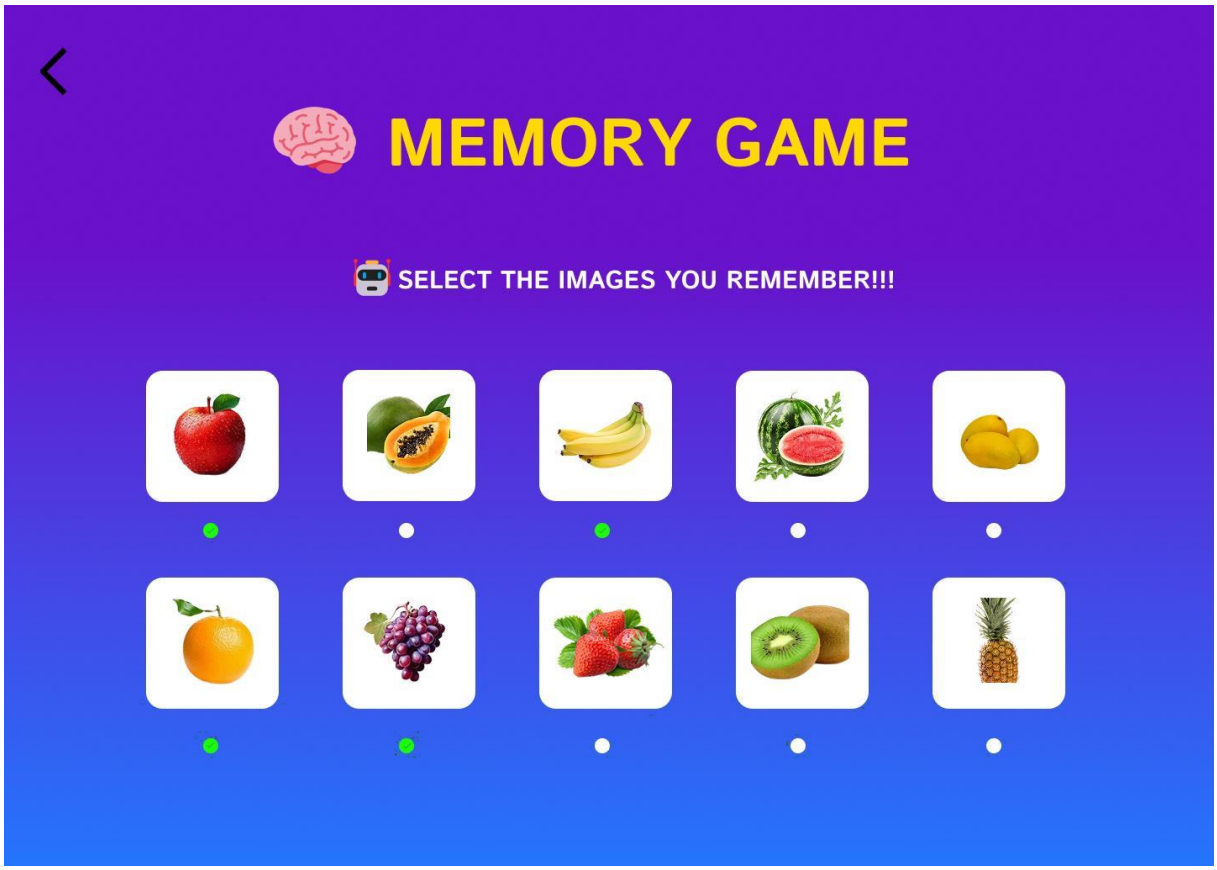
## User Interaction Flow

- The user clicks on the **small circular button below each image**.
- Each click is checked immediately by the system.
- There are two possible outcomes:

## Correct Selection Behaviour

- When the user selects a **correct image**:
  - The circular indicator turns **green**
  - The image is marked as correctly remembered
- The user can continue selecting more images
- If **all correct images are selected without mistakes**:
  - The system shows a **success feedback screen**
  - Message displayed:
    **"Excellent Memory!"**

✓ This creates positive reinforcement
✓ Encourages accuracy and attention
✓ No unnecessary confirmation steps



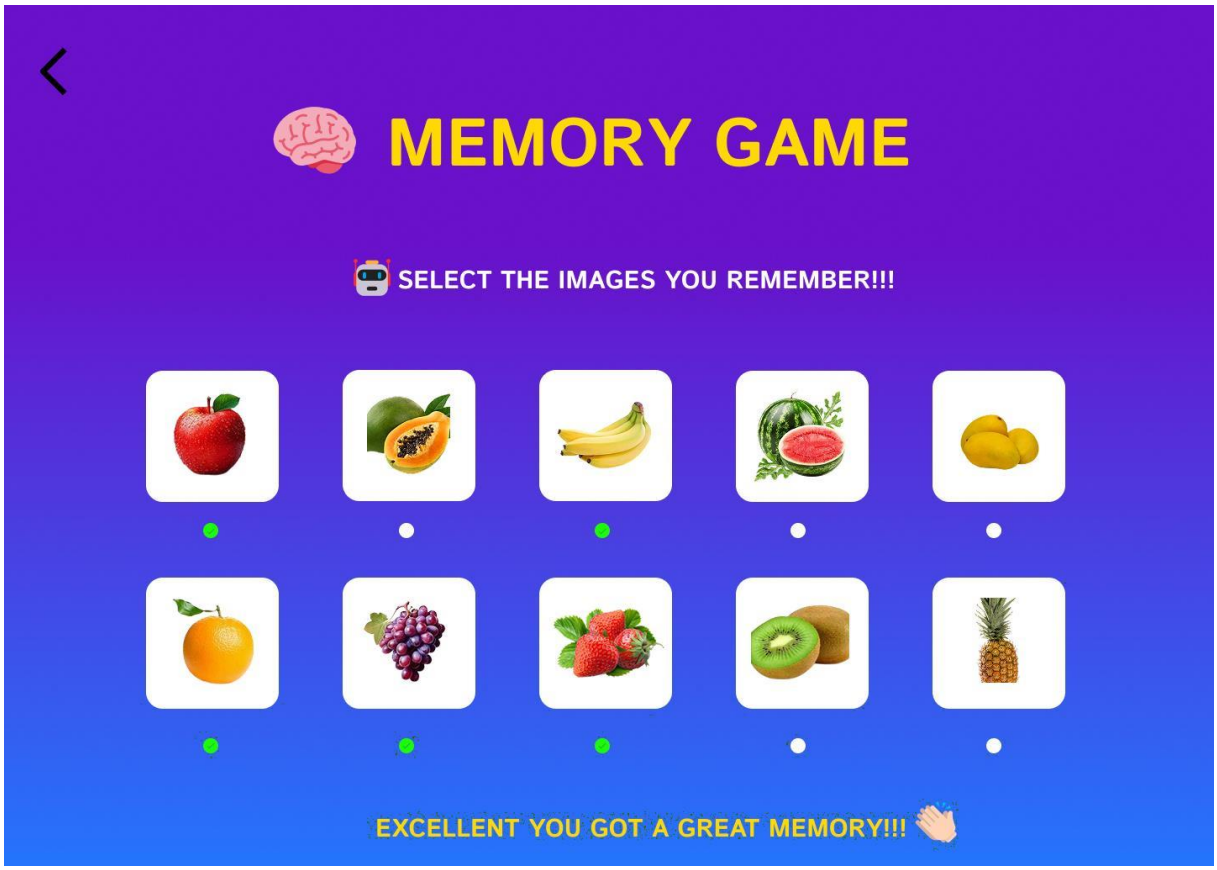

### Wrong Selection Behaviour

- If the user selects **any wrong image**:
    - The selected wrong image turns **red**
    - All correct images are highlighted in **green**
    - The mistake is shown immediately

There is **no second chance** once a wrong image is clicked.

## FRAME 4: RESULT / FEEDBACK SCREEN

### If User Selects All Correct Images



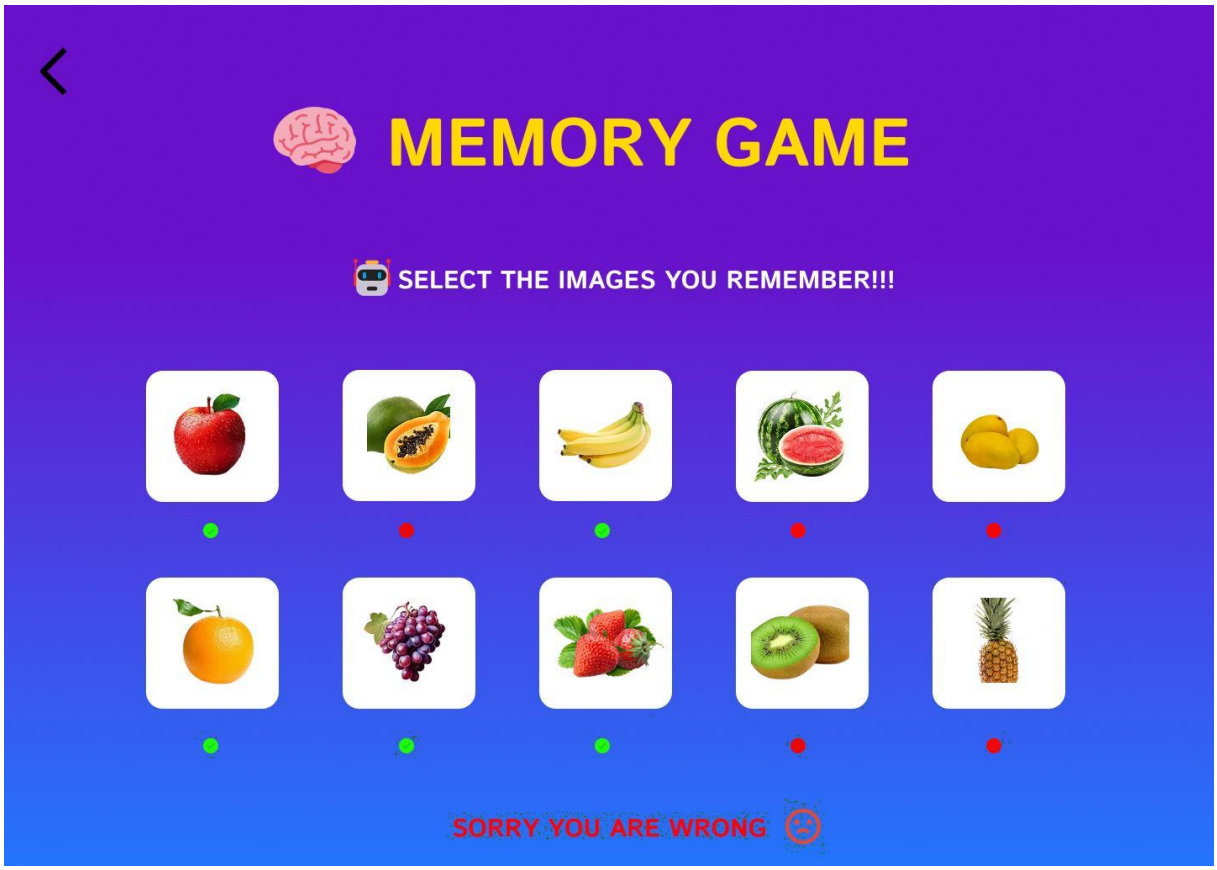

- Feedback Message:

**“Excellent You Got A Memory!”**

- Visual Feedback:
    - All correct images highlighted in **green**
- Action Button:
    - **Try Again**

This allows the user to replay and improve memory performance

**If User Clicks a Wrong Image**

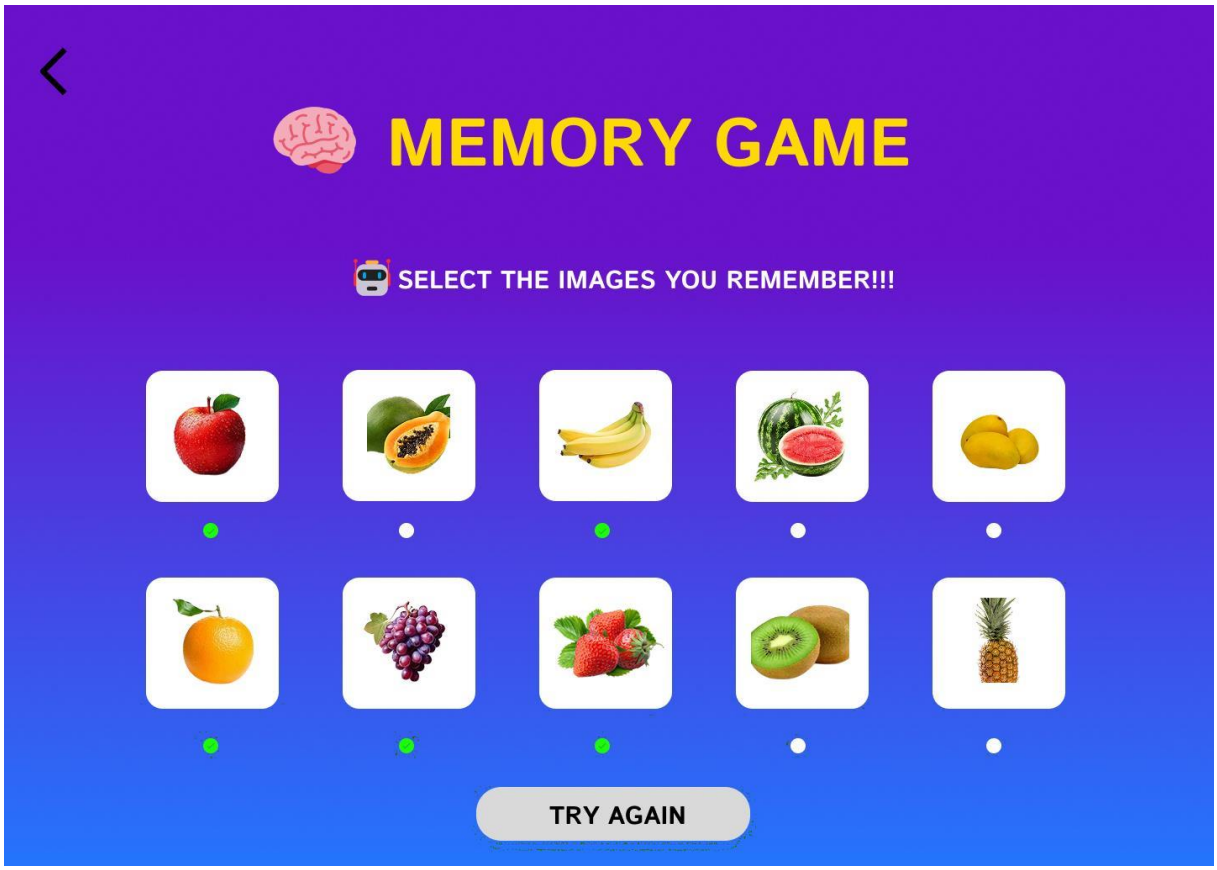


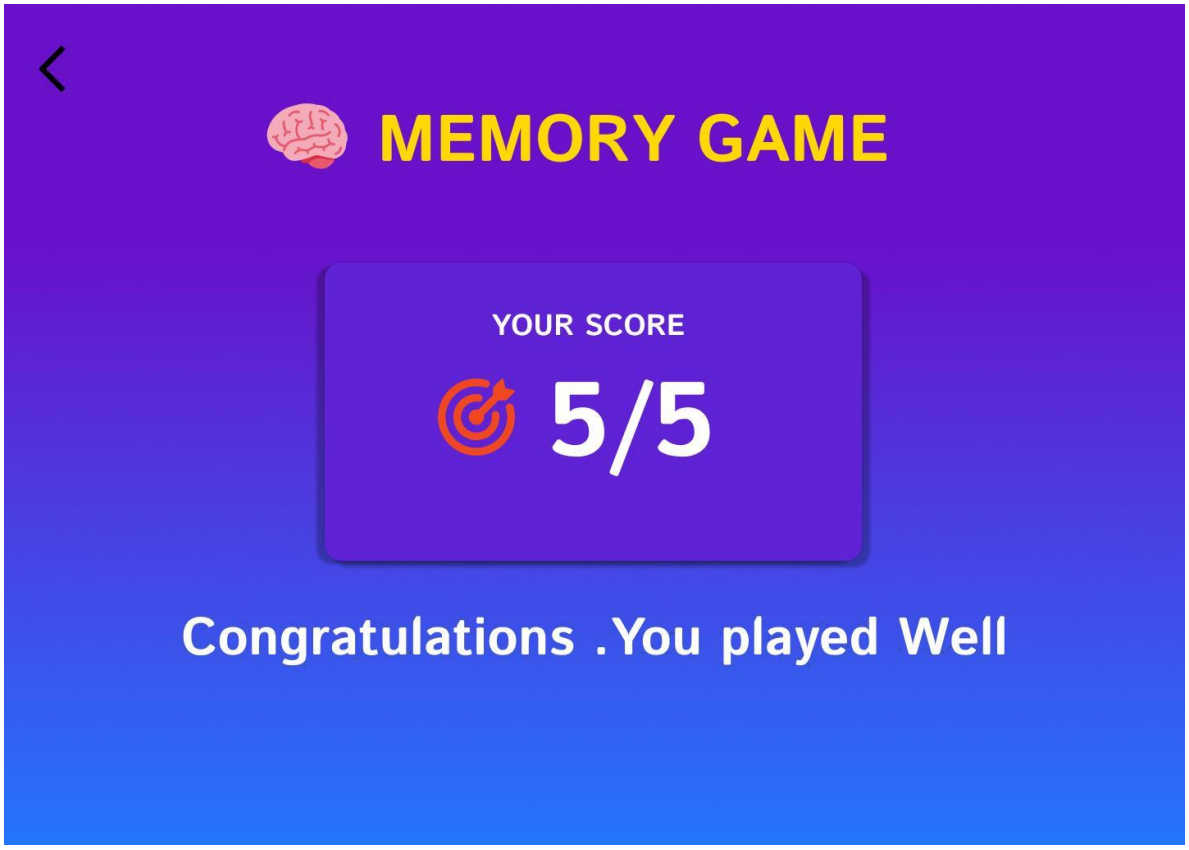

- Feedback Message:

**“Sorry, you are wrong.”**

- Visual Feedback:
    - Correct images → **Green**
    - Wrong image → **Red**
- Action Button:
    - **Try Again**

The user must restart the game.


MEMORY GAME
SELECT THE IMAGES YOU REMEMBER!!!
TRY AGAIN


MEMORY GAME
YOUR SCORE
5/5
Congratulations .You played Well

**PROTOTYPE LINK:**

https://www.figma.com/proto/jFJcWMQfy0jPyk8tPOTKLe/experiment2?node-id=7-20&t=Op1QO5aKs9dfLXBn-1&starting-point-node-id=7%3A20

## Key Features of the Memory Game Design

### 1. Visual Memory-Based Gameplay

Users are shown a set of images and must recall them accurately, testing short-term memory and concentration.

### 2. Instant Selection Feedback

As soon as the user clicks an image:

- Correct choice → highlighted in **green**
- Wrong choice → highlighted in **red**

No submit button is required.

### 3. Real-Time Result Evaluation

- If **all selected images are correct**, the system shows **"Excellent Memory"**
- If **any wrong image is selected**, the system immediately shows **"Sorry, you are wrong"**

### 4. Retry-Based Interaction

After result display:

- Only **"Try Again"** option is shown
- User can replay without reloading the page

### 5. No Manual Submission

The game logic works automatically based on user interaction, making it faster and more engaging.